

Original papers

A novel green apple segmentation algorithm based on ensemble U-Net under complex orchard environment

Qianwen Li^a, Weikuan Jia^{a,b,*}, Meili Sun^a, Sujuan Hou^{a,b}, Yuanjie Zheng^{a,b}^a School of Information Science and Engineering, Shandong Normal University, Jinan 250358, China^b Shandong Provincial Key Laboratory for Novel Distributed Computer Software Technology, Shandong Normal University, Jinan 250358, China

ARTICLE INFO

Keywords:

Green apple
Fruit segmentation target
U-Net model
Edge structure
ASPP

ABSTRACT

Fruit segmentation is a critical step for fruit recognition, and key to the efficiency of a robotic vision system for harvesting and accurate orchard yield estimation. Due to the unstructured characteristics of orchard environment, the segmentation performance of traditional segmentation algorithms is insufficient for handling green target fruits in a complex environment. Deep learning algorithms bring a fresh perspective to target segmentation. This study proposes an ensemble U-Net segmentation model suitable for small sample datasets. Edge structures are designed by integrating residual blocks and gated convolutions to obtain the boundary semantic information of the target image; atrous convolutions are applied to resolve the contradiction between the resolution of the feature map and the receiving field, retain more multiscale context information and achieve target fruit segmentation. An atrous spatial pyramid pooling (ASPP) structure is applied to merge the edge features and the high-level features of U-Net. The experimental results show that the proposed method effectively improves the segmentation accuracy of the target fruit and the generalization ability of the model. The proposed method extends the application scope of the harvesting robot and orchard yield measurements, thereby providing a theoretical reference for other fruit and vegetable target fruit segmentation efforts.

1. Introduction

Early apple yield estimation and harvesting operations are important parts of the apple production process. An early yield estimation can provide a decision basis for analytical management of apple production. Automated harvesting guarantees real-time apple production harvesting (Bac et al., 2014; Jia et al., 2020a). Both of these operations are labor- and time-intensive. Currently, despite the advancement of automation technologies, manual work is still the main workforce in producing apples. However, such labor intensity and work efficiency are low. With the gradual reduction of agricultural labor force, these factors have caused unprecedented disruptions to the production of apples. Therefore, there is an urgent need to carry out research on automated operations to improve the accuracy and efficiency of apple production operations. The emergence of the field of machine vision has added new vitality to agricultural production, transforming the modes of agricultural operations for improved agricultural productivity (Diego et al., 2018; Tripathi and Maktedar, 2020). Vision systems have brought new opportunities to early yield estimation and automated harvesting of apples, achieving many successes in target image segmentation (Wang

et al., 2019; Xu et al., 2019), target fruit recognition (Liu et al., 2018c), and harvesting path planning (Davidson et al., 2016; Silwal et al., 2017).

Image segmentation is the basic premise of target recognition (Sui et al., 2017; Majeed et al., 2020). The effect of segmentation is directly related to the accuracy of target recognition. Early apple yield estimation and automated harvesting require accurate recognition of target fruits. Therefore, effective segmentation of apples in images becomes a key factor for the success of early apple yield estimation and automated harvesting. In recent years, research on apple image segmentation has attracted attention from many scholars and has resulted in a series of publications (Bargoti and Underwood, 2017a,b; Li et al., 2019; Lv et al., 2019a; Jia et al., 2020b). For apples with an uneven skin color, Liu proposed an apple segmentation method based on superpixel features. A simple linear iterative clustering algorithm was applied to segment the target image into several superpixel units. Then, texture and color features were extracted, and a support vector machine (SVM) classifier was designed to distinguish the target fruit and the background with an accuracy of 0.9214 (Liu et al., 2019). Wang presented an overlapping target fruit segmentation algorithm based on the fusion of K-means and normalized cut (Ncut) algorithm for overlapping apple fruit targets. A

* Corresponding author.

E-mail address: wkjia@sdnu.edu.cn (W. Jia).<https://doi.org/10.1016/j.compag.2020.105900>

Received 23 June 2020; Received in revised form 12 November 2020; Accepted 14 November 2020

Available online 3 December 2020

0168-1699/© 2020 Elsevier B.V. All rights reserved.

spline interpolation algorithm was used to reconstruct the contour of occluded apple targets, and segmentation errors were significantly reduced (Wang et al., 2015). Ahmad designed a robust segmentation algorithm for apple images based on a fuzzy inference system and used fuzzy c-means to segment different colored target fruits during the growth stage of the apple. The algorithm improved the generalization ability of the segmentation method (Ahmad et al., 2018). Ji applied an apple image segmentation algorithm based on growing regions and color features and also designed an SVM algorithm for apple recognition to improve the accuracy and efficiency of target fruit recognition. The recognition success rate of this method is about 89%, and the average recognition time is 352 ms (Ji et al., 2012). For bagged green apples, Lv presented an image segmentation method that was used after extracting the normal light regions and highlighted regions of the fruits. The contrast limited adaptive histogram equalization (CLAHE) algorithm was adopted to improve edge definition in an image; an Otsu segmentation algorithm was used to extract the normal light region of a fruit. This new method achieved efficient segmentation of the target fruits (Lv et al., 2019b). The abovementioned methods obtained ideal segmentation results under certain conditions because some of the models had relatively strict conditions when learning the target features. In the actual complex environment of the orchard, these algorithms are not applicable.

With the rapid development of deep learning theory and advances in hardware equipment, many computer vision problems have been solved using deep network architecture. End-to-end processing methods using deep neural networks are now more widely used for image processing, and have achieved relatively promising results in image segmentation (Guo et al., 2018; Liu et al., 2018a; Zhang et al., 2020). Deep learning and intelligent optimization algorithms bring a new perspective to fruit segmentation targeting (Kuruvilla et al., 2016; Hou et al., 2018; Liu et al., 2018b), thereby attracting the attention of scholars in the agricultural field. Kang proposed the DaSNet-V2 network structure, in which visual sensors are applied to real-time detection and segmentation of apples in orchards and branches are segmented by semantic segmentation. Visual sensors were used in field trials, and the results showed that the new method was efficient and reliable (Kang and Chen, 2020). Jia presented a target fruit recognition algorithm using optimized Mask R-CNN for overlapping apple targets. Residual neural network (ResNet), combined with dense convolutional network (DenseNet), was used as the backbone network for feature extraction. The region proposal network (RPN) structure was designed to generate a region of interest (ROI), and the new method improved the recognition accuracy for overlapping target fruit (Jia et al., 2020c). Chen proposed a blob detector based on a fully connected CNN to extract candidate regions from images; the object regions were segmented, and then, a subsequent CNN counting algorithm was used to calculate the number of fruits (Chen et al., 2017a,b). Bargoti used a multiscale multilayer perceptron and a CNN to segment apple images and extract the apple target from the images, and the detection accuracy is above 0.9. The method obtained a high detection accuracy but could not recognize all the fruits appearing in the cluster (Bargoti and Underwood, 2017a,b). The successful application of deep learning to other fruits and vegetables also provided a theoretical reference for apple target segmentation (Ji et al., 2019; Koirala et al., 2019a; Koirala et al., 2019b). These gratifying achievements have promoted research on apple target fruit recognition. However, for unstructured apple orchards, which are affected by natural light, weather, green fruits, difficult angles of acquisition, a large number of samples, etc., the collecting images were more intricate, leading to new challenges in the design of deep network architectures.

For green apple images in complex orchard environments, this study proposes a novel ensemble U-Net segmentation model with small sample datasets. Firstly, the U-Net structure is used to obtain high-level semantic information. Secondly, based on the U-Net model, an Edge structure is developed by integrating the residual blocks and the gated convolutional layer (GCL) to obtain the boundary semantic information

of the image. Finally, to retain more scale context information and achieve efficient segmentation of the target images, the spatial pyramid structure of the atrous convolution is introduced to merge with high-level semantic features of U-Net and the edge features of Edge.

The structure of this paper is as follows. In section 2, image acquisition and dataset construction are described. Section 3 introduces the improved structure of the model, illuminating how to optimize the model and the specific implementation steps of the integrated Edge structure. In section 4, the experiment is described, including the experimental design and analysis of the result.

2. Data acquisition

Datasets are the foundations of model learning processes, especially for deep learning models. Image acquisition is the first step of image segmentation. After the acquisition step, the images need to be pre-processed. Image resolution, image quality, image samples and the number of samples are crucial factors in determining whether a good deep learning model can be trained. This work takes apple images under complex circumstances as the research object. To obtain the target apple images as comprehensive as possible, during data collection, the actual environment of the orchards is fully considered. Having a rich dataset can avoid overfitting in the model training, thereby improving the generalization ability of the model.

2.1. Images acquisition

Below are some details of the data acquisition conditions:

- *Image acquisition location:* Longwangshan apple production base in the Fushan District, Yantai City, Shandong Province (the agricultural information technology experimental base of Shandong Normal University).
- *Image acquisition device:* a Canon EOS (electro-optical system) 80D SLR (single-lens reflex) camera. This camera has a CMOS (complementary metal oxide semiconductor) image sensor. The image resolution is 6000×4000 pixels, with RGB color images stored in the JPG format.
- *Image acquisition environment:* To simulate the actual working scene of a green apple harvesting robot, the camera lens direction was set perpendicular to the ground. The shooting distance of some images is the same as the average length of the picking robot arm, which is 40–60 cm, and the shooting distance of the other images is slightly farther to provide samples with smaller targets. We capture from different angles around different fruit trees. The image collection process has two sessions, i.e. daytime session and nighttime session, and the images were acquired under natural illumination conditions (including conditions of direct sunlight and back-lighting) in the daytime and under light-emitting diodes (LEDs) at night. Numerous apple images were captured from multiple directions, including occlusion by branches and leaves, overlapping fruits and single fruits, etc.
- *Apple category:* Gala.

In the actual processes of the robot harvesting, the speed of recognition is one main performance metric of real-time apple fruit recognition. Images may be blurred during real-time shooting, and low resolution may be disadvantageous for target recognition. To make the trained algorithm better for target recognition at low resolutions and to reduce the running time of subsequent experiments, reducing the resolution of the collected images is considered. In addition, the U-Net structure is more suitable for an image with a size of 512×512 , and the size of the input image is one of the important factors affecting the U-Net segmentation effect. If the original image resolution is directly reduced, the target apple in the image will be deformed. In order to avoid this situation, the original images are cropped to the size of 4000×4000

pixels and further downsampled to 512×512 pixels. Some sample images are shown in Fig. 1. In Fig. 1, the actual images of apples in various environments such as daytime, night, occlusion, overlap, back-lighting, long shot and after rain are shown.

2.2. Dataset production

To ensure that the apple harvesting robot works around the clock to adapt to the actual harvesting scene, a variety of conditions of the apples have been considered, including the facts that 1) occlusions can make the shape features incomplete, and 2) the brightness level of light can affect the color intensity of an apple to vary considerably. The dataset was enriched as much as possible to ensure the completeness of various apple images.

A total of 211 apple images were uniformly selected from the collected images in different environment images. Then the labeling software was used to manually label and store the images. Considering that the insufficient sample number may cause overfitting when a model is trained, data augmentation was adopted to prevent overfitting. Based on the actual harvesting scene, the original image was processed with image enhancement and reductions of brightness, contrast, chroma, and sharpness to improve the generalization capability of the trained model. The brightness, contrast, and chroma of the images were enhanced by 1.5 times compared to the original image; the sharpness was enhanced by twice compared to the original images, and the brightness, contrast, chroma, and sharpness were reduced by 70%, 70%, 70% and 20%,

respectively. The effect of data augmentation of the original image are illustrated in Fig. 2.

After data augmentation, the original annotations are still valid. In this study, 1899 apple images in total are obtained. Table 1 shows the details of the dataset in terms of the number of images in each specific environment more clearly.

After completing the preparation of apple image acquisition, data augmentation and annotation, the apple data set are divided according to the common division principles of small data sets. Among them, 60% of the apple images are used as the training set, 20% as the validation set, and 20% as the test set. There is no overlap between the three subsets. There are apple images in different environments in each subset, and the specific division details are shown in Table 2.

3. An ensemble U-Net segmentation algorithm

The lack of large apple datasets is a tremendous challenge for apple harvesting robots in the recognizing apples. Due to the commonalities of the object recognition tasks, some models can solve the problem of insufficient training samples for specific image recognition tasks through parameter migration. However, most pretrained CNN networks are deep and wide, and transfer learning relies on a pretrained model to complete feature extraction. To a certain extent, this process determines the quantity of the calculations and parameters of the model during transfer learning. Although a recognition model based on transfer learning can reach high accuracy, it is difficult to implement a low-end

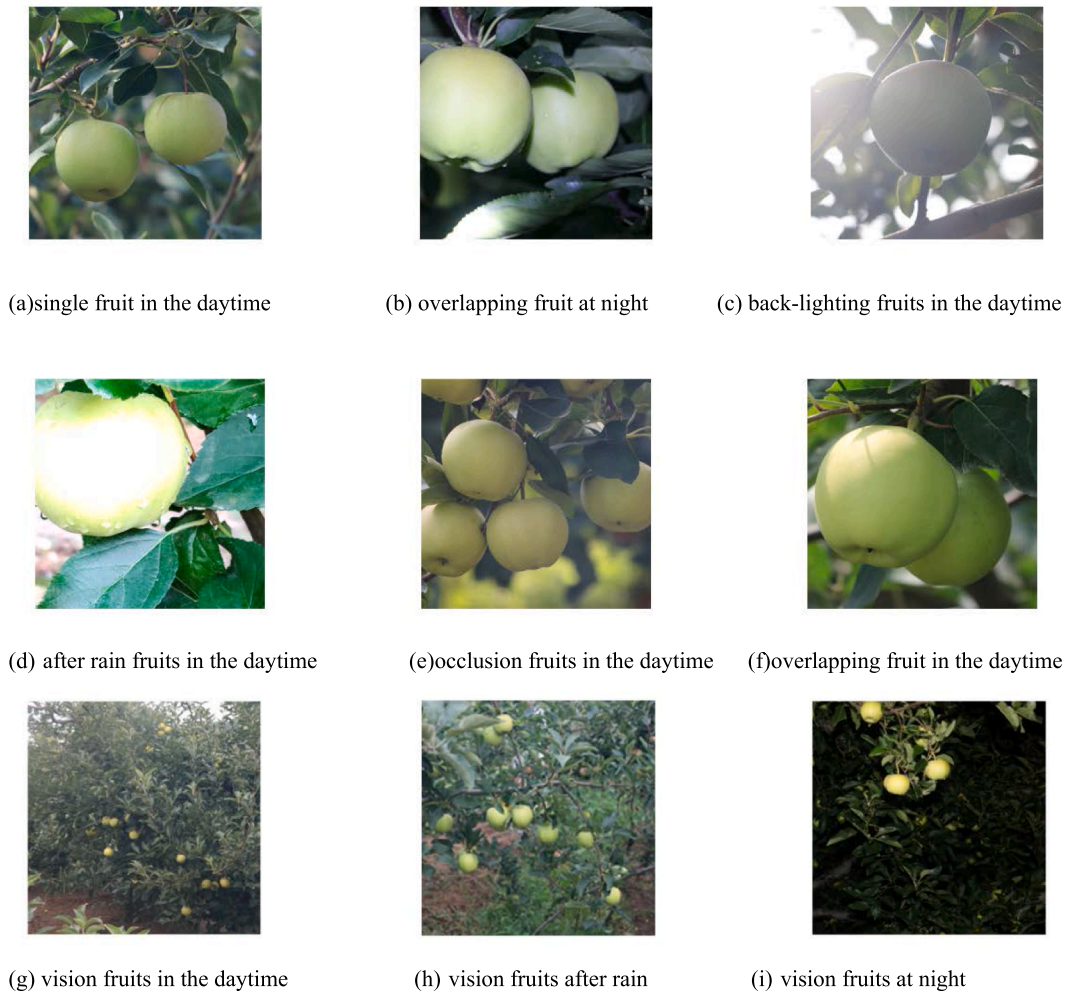


Fig. 1. Apples images under the complex orchard environment.



Fig. 2. The enhancement effect of apple image.

Table 1
Number of apple images per specific environment.

Specific environment	Covered fruit	Overlapping fruit	Single fruit
Day	398	351	298
Night	325	297	230

Table 2
The division of the data set.

Division situation	Covered fruit	Overlapping fruit	Single fruit
Training set	433	388	316
Validation set	145	130	106
Test set	145	130	106

configuration model deployment due to the large number of model parameters and the long reasoning time. The U-Net model is suitable for small sample datasets, and the structure of the model is simple, with high-quality segmentation performance. Therefore, this paper uses U-Net as the base network. Based on the characteristics of apples, the U-Net model is improved and applied to the segmentation of the target fruit.

3.1. U-Net architecture

U-Net is a popular image semantic segmentation algorithm (Ronneberger et al., 2015), that was initially used in the medical field and later widely used in other fields. The U-Net's structure, as an encoder-decoder structure, is similar to a large 'U' letter. The entire structure of the U-Net network is shown in Fig. 3. First, Conv + Pooling for downsampling is

developed to extract image features. Second, deconvolution for upsampling is used to recover an image size. Then the feature map is obtained by upsampling, and the underlying feature map before cropping is fused. Finally, upsampling is applied again. This process is repeated until the output feature map is obtained. The output segmentation map is obtained through optimizing the softmax loss function.

The energy function is defined as a pixel-wise soft-max of the final feature map combined with a cross entropy loss function. The soft-max is defined as:

$$p_k = \exp(a_k(x)) / (\sum_{k=1}^K \exp(a_k(x))) \quad (1)$$

where $a_k(x)$ denotes the activation in feature channel k at each pixel position, k is the number of classes and $p_k(x)$ is the approximated maximum-function.

The weighted penalty formula is defined as:

$$E = \sum \omega(x) \log(p_{l(x)}(x)) \quad (2)$$

where $l(x)$ is the true label of each pixel and ω is a weight map.

The weight map is then computed as

$$\omega(x) = \omega_c(x) + \omega_0 \cdot \exp\left(-\frac{(d_1(x) + d_2(x))^2}{2\sigma^2}\right) \quad (3)$$

where ω_c is the weight map used to balance the class frequencies, d_1 denotes the distance to the border of the nearest cell and d_2 is the distance to the border of the second nearest cell.

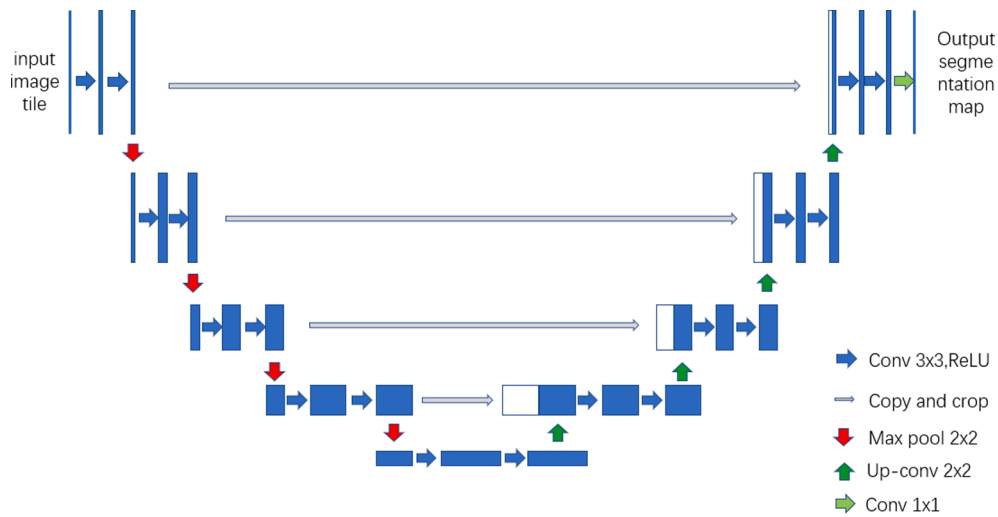


Fig. 3. U-Net architecture.

3.2. An improved U-Net model

An improved U-Net model is proposed and applied to the apple harvesting robots. Usually, a clean background can help to achieve semantic image segmentation. However, the background in the actual orchard harvesting scene are often intricate, and this complexity also greatly increases the difficulty of image segmentation. There are many types of apples, among which green apple is mainly considered in this paper. It is challenging to perform segmentation because the green apples are similar as the background branches and leaves in terms of color (Rakun et al., 2011; Sun et al., 2019). In addition, the boundary of the apples tend to be blurred.

Focusing on the problem of the blurring of the segmentation boundary, this paper adds and integrates an Edge structure on the basis of U-Net. This structure includes a residual block and a gated convolution, which can effectively extract the relevant information of the target fruit boundary and focus on the processing of detailed information. The U-Net basic network is used to extract the high-level features of the input apple image and then use the Edge structure to extract the shallow

features of the apple image. Atrous convolution is applied to solve the contradiction between the resolution of the feature map and the receiving field. To retain more multiscale context information and achieve target fruit segmentation, the spatial pyramid structure of the atrous convolution is developed to merge the edge features of Edge and the high-level features of U-Net. The structure diagram of the model is shown in Fig. 4.

The ensemble U-net algorithm is applied to the vision system of the apple harvesting robot to enable it to accurately recognize the target apple.

3.3. Gated convolutional layer (GCL)

The quality of apple image segmentation is closely related to the semantic boundary. In the green apple images, the color representations of the green apple target and the background branches and leaves are similar. There is a problem of boundary blur in the segmentation. This study introduces a GCL (Takikawa et al., 2019) to solve the problem of unclear segmentation boundaries by reducing the gradient propagation.

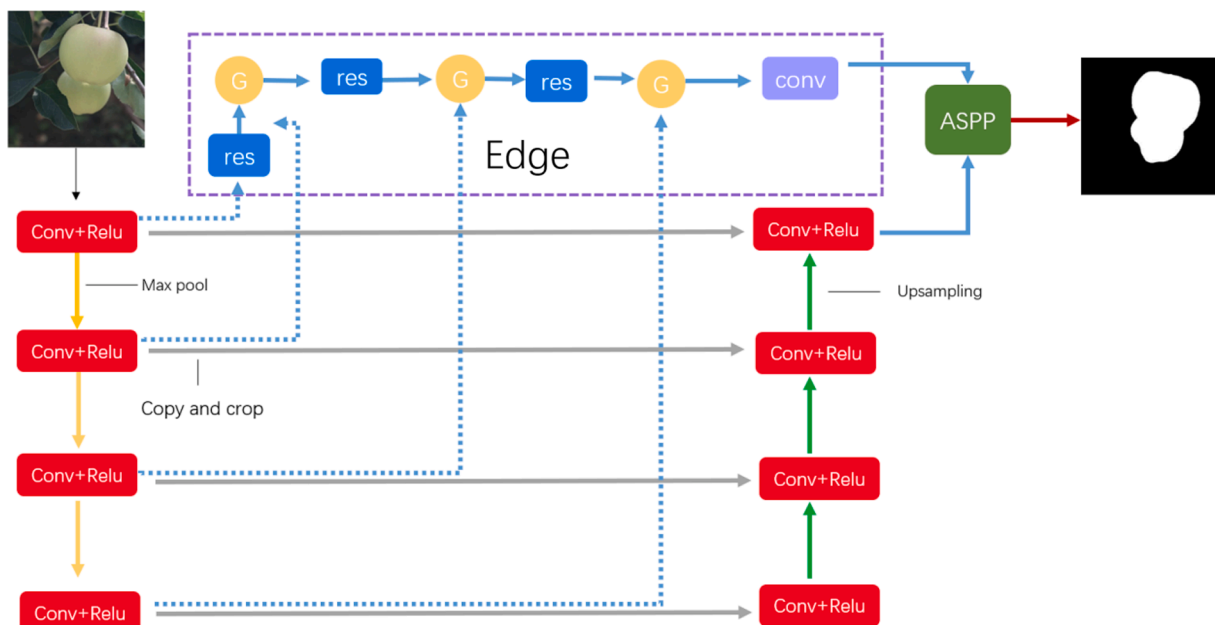


Fig. 4. Ensemble U-Net model based edge structure and ASPP.

GCL is the core component of the Edge structure and helps the Edge structure process only the relevant boundary-related information of the apple by filtering out the rest of the information. Therefore, it can also be considered as an attention mechanism. In this study, it can be considered as a collaboration between shallow structures and deep structures, and it helps Edge to focus on only the relevant parts from the beginning, thereby enabling GCL to adopt an effective shallow structure that processes the image at a very high resolution.

GCL is used many times in the model. GCL lets the Edge structure focus on the boundary information. The attention map formula is as follows:

$$\alpha_t = \sigma(C_{1 \times 1}(e_t || c_t)) \quad (4)$$

where e_t is the output of Edge, c_t is the output of the convolution, $||$ means to concatenate the two outputs, $C_{1 \times 1}$ means to convolve the concatenating result by 1×1 , and σ is the sigmoid function. First, the output of the convolution and the output of Edge are concatenated; then the splicing result is convolved by 1×1 . Finally, the sigmoid function is used to form an attention map α_t .

A new map is obtained through a GCL layer, and the formula of the map is as follows:

$$E_t^{(ij)} = (e_t * \omega_t)_{(ij)} = ((e_{t(ij)} * \alpha_{t(ij)}) + e_{t(ij)})^T \omega_t \quad (5)$$

The Edge map obtained from the previous layer, is pixel-wise dot-multiplied with the obtained attention map α_t to obtain the ROI information. The new Edge map is the product of the transposition of the ROI information, the previous Edge map and the weight kernel of the channel. The whole process can be regarded as the process of an attention mechanism.

3.4. Edge structure

Currently most advanced deep CNN image segmentation algorithms process color, shape and texture information together. The results are not ideal because they contain very different information related to recognition. The model proposed in this paper treats different information separately, and the Edge structure focuses only on extracting shallow information, such as image texture, shape, and edge. The Edge structure consists of GCL and a few residual blocks. The GCL makes Edge focus on the boundary information.

Deep CNN have a very strong ability to characterize features. When dealing with complex tasks, it is necessary to design a deeper model. But due to the disappearance of gradients, explosions often occur in the training of deep networks. In addition, the degradation problems in deep networks makes the accuracy of the network become saturated or even decrease when the network level increases, such that training deep convolutional neural networks becomes more difficult. The appearance of residual blocks is used to solve the problem of performance degradation due to the increased depth. In theory, the training effectiveness continuously increases as the depth of the neural network increases. However, in fact, due to the difficulty of training, deeper networks will be more likely to cause degradation problems. Thus the effect is not as good as that for relatively shallow networks. The residual block solves

this problem effectively by using multiple layers with parameters to learn the representation of the residual between the input and output. This significantly improves the training speed and prediction accuracy of the deep network.

In this paper, the Edge structure takes the convolution and the output of this structure as the input, and the features of the shallow convolution are input into the residual block of the weighted value ω_t . Then, the boundary related information is extracted through the GCL. Finally, the real image is used. The Edge structure monitors the semantic boundary to produce the final boundary-related information. The Edge structure is shown in Fig. 5.

For the segmentation task, the Edge structure uses the loss of the obtained boundary result and the real boundary result to supervise the boundary segmentation effect. The loss function can be written as follows:

$$L^{\theta, \varphi, \gamma} = \lambda_1 L_{BCE}^{\theta, \varphi}(e, E) \quad (6)$$

where binary cross entropy (BCE) is used to predict the boundary information and supervise the updated Edge parameters. λ_1 is the hyper-parameter that controls the loss weight. e is the real boundary, and E is the predicted boundary value.

To prevent the model from overfitting, a regularization term is added to the loss function. The output represents whether a pixel belongs to a certain category of a picture. It can be calculated by taking a spatial reciprocal on segmentation output:

$$\zeta = \frac{1}{\sqrt{2}} ||\nabla(G * \arg\max_p(y^k | r, s))|| \quad (7)$$

where G denotes the Gaussian filter and $p(y^k | r, s)$ denotes a categorical distribution output of the fusion module.

3.5. Atrous spatial pyramid pooling (ASPP)

The sizes of the feature map output by the Edge structure and the U-Net basic network are different. If the images are directly resized to the same dimension to be merged, some features will be distorted or disappear. This study introduces the atrous spatial pyramid pooling (ASPP) (Chen et al., 2017a,b) and uses ASPP to fuse the Edge structure and the U-Net basic network output feature map to retain more multi-scale context information.

The atrous convolution is introduced in the ASPP structure, which is mainly used for image segmentation. In the traditional CNN model, downsampling to expand the receptive field, i.e. abstracting feature information, is very helpful for high-level image processing tasks, such as image classification tasks. However, to output a sum for the segmentation task of a predictive image input with the same resolution, the effect is not as desired. Without the use of pooling and downsampling, the atrous convolution plays the same role of increasing the receptive field so that each output of the convolution has a larger range of information, and the atrous convolution solves the contradiction between the feature map resolution and reception. A spatial pyramid structure with an atrous convolution has a wide range of applications in image segmentation, and its architecture is shown in Fig. 6.

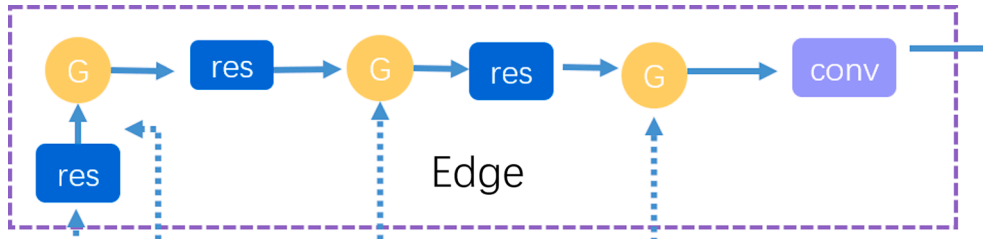


Fig. 5. Edge structure architecture.

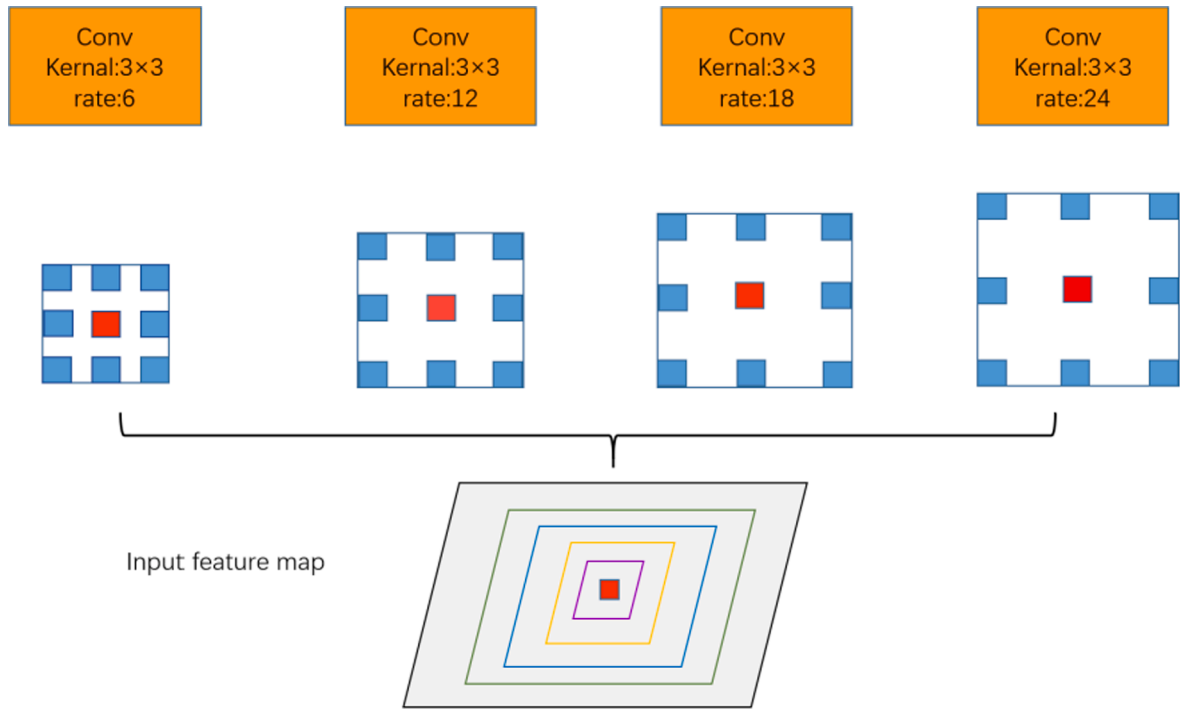


Fig. 6. ASPP architecture.

ASPP generates input pictures with multiple scales. In this paper, the input picture is a feature map of different sizes output by the Edge structure and the U-Net basic network. These feature maps employ atrous convolutional layers with different sampling rates (such as sampling rates of 6, 12, 18 and 24); parallel sampling of these feature maps is equivalent to capturing the context of the image at multiple scales. In short, ASPP is used by integrating the edge features that the Edge structure extracted and the high-level features of U-Net to retain more multiscale context information, which further strengthens the segmentation effect and obtains better effects of the apple segmentation map.

4. Experiment

4.1. Experimental platform and model training

This experiment is carried out on a laboratory server using a PyTorch-based deep learning framework. The operating system is 64-bit Windows 10 and the workstation has an Intel i7-10510U CPU (16.00 GB memory) and NVIDIA GeForce MX250 GPU. The code is developed using Python.

This experiment uses the stochastic gradient descent method to train the network in an end-to-end joint manner. The initial settings of model parameters refer to existing model application examples, and then fine-tune them. The weight decay rate is set to 0.005; the kinetic energy factor to 0.9; the maximum number of iterations to 3500; the starting learning rate to 0.001, and the decay 10 times after each iteration 500 times. In the training model, it is set to output the model every 100 times. The segmentation results of the training set and the validation set are very close. Excessive training may lead to overfitting, so it is necessary to test and evaluate the trained model.

4.2. Evaluation criterion

In this paper, the evaluation criterion in the segmentation process mainly uses Dice Similarity Coefficient (DSC). Dice coefficient is a measure index of set similarity, which is one of the commonly used evaluation indices in image segmentation. Usually it is used to calculate

the similarity of two samples. And the value range is 0–1, while 1 indicates the best segmentation result, and 0 the worst. In order to evaluate the results of apple segmentation more objectively, this study also considers the use of recall and precision as evaluation indicators.

The expressions of Dice, Recall, Precision are as follows:

$$Dice = \frac{2TP}{FP + 2TP + FN} \quad (8)$$

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

$$Precision = \frac{TP}{TP + FP} \quad (10)$$

Among them, according to the relationship between the predicted value and the true value, the apple sample can be divided into the four types as follows:

- true positive (TP): The sample is judged as a positive sample, and in fact it is also a positive sample;
- false positive (FP): The sample is judged as a positive sample, but in fact it is a negative sample;
- true negative (TN): The sample is judged as a negative, and in fact it is also a negative sample;
- false negative (FN): The sample is judged as a negative, but in fact it is a positive sample.

4.3. Result evaluation

4.3.1. Apple target segmentation result

In order to simulate the real picking scene in the orchard, the improved U-Net apple image segmentation algorithm is used to process the apple images. For different apple images of real scenes, including overlapping fruits, occlusions of branches and leaves, night environment, long shot, etc., multiple apple segmentation effect pictures are selected accordingly for analysis. The graph of apple segmentation effect is shown in Fig. 7.

It can be seen, from Fig. 7, that the perspective of segmentation



Fig. 7. Segmentation effect of the method.

effects, single apples, occluded apples, and overlapped apples can be segmented better from the complex apple images. Especially the segmentation edges of occluded apples and overlapped apples are relatively clear. Compared with the accurate segmentation of the close-range image, apple segmentation for the distant images has also achieved promising results, where apple boundaries are clearly and accurately identified. The situation of the long-range images is more complicated as the apple targets appear smaller, hence increasing the difficulty level for precise segmentation.

In order to more clearly describe the effect of the segmentation method proposed in this paper, the target fruit recognition effects of three different growth postures (overlapping fruit, occlusion fruit, independent fruit) under complex environment were statistically analyzed, as shown in Table 3.

4.3.2. Comparison with other methods

To further analyze the performance of the network, in this paper, the ensemble network is compared with U-Net, Deeplab V3 +(Chen et al., 2018), and PSPNet (Zhao et al., 2017). Among these, U-Net is the basic network. The experimental results are shown in Fig. 8.

It can be seen from Fig. 8 that, in cases of unobstructed apples regardless it is a long-range shooting or close-range shooting, the segmentation difference of these algorithms is not very noticeable. The edges of the apples can be segmented well. However, as mentioned, overlaps, occlusions and adhesions of apples, and the closeness between the fruit skin color and the branches and leaves in the background are common in the real-world environment. This makes image recognition more difficult, especially the precise recognition of edges. Apple images obtained from distant shots are more complex than close-range images, hence also more difficult to recognize. Analyzing and comparing the segmentation results of the algorithm, in this paper, the contours of the occlusion apples and overlapping fruits are relatively clear under complex orchard environments. Our method clearly demonstrates better segmentation results than other methods.

It can be derived from Table 4 that the network model can effectively realize the recognition of apple images. The recognition rates of single fruits, covered fruits and overlapping fruits are 97.15%, 93.58% and 94.46% respectively, and the comprehensive recognition rate reaches 95.11%. Through the above analysis, despite some misidentification and inaccuracy in recognition, it still clearly shows that the proposed method can more accurately identify the apple targets in the images compared with other methods, and the edge segmentation clearly outperforms other methods too. In addition to evaluating the recognition rate of the algorithm, it is also necessary to consider the computational time for the

algorithm. It can be seen from Table 5 that the average time for recognizing the targets is 0.39 s, and the image recognition speed is relatively faster compared with other methods. After the above analysis, the algorithm has a strong anti-interference ability and generalization ability.

4.3.3. Comparison with other data sets

In order to further verify the performance of the network, the open standard data set is employed on our network for testing. Three data sets are selected: Pascal Voc 2012, Cityscapes, and Adobe's Portrait Segmentation for comparison with ours. The two data sets of Pascal Voc 2012 and Cityscapes are relatively complex, while the Adobe's Portrait Segmentation data set is slightly simpler. We tested the four data sets with different methods, and the results are shown in Table 6. Three evaluation indicators show that the results of Pascal Voc 2012 and Cityscapes are slightly worse than our data set, and Adobe's Portrait Segmentation has a better segmentation accuracy. From the comprehensive results, our method has a better segmentation performance.

5. Conclusion

In order to avoid the influence of lighting conditions, overlapping and other occlusions on image recognition and to use the characteristics of strong feature extraction abilities of the CNN, this paper proposes an apple segmentation algorithm based on deep learning, which is applied to apple harvesting robots. The ensemble U-Net convolutional neural network is used as the base network architecture, and its structure and parameters are optimized and improved to better extract image features. The accuracy of the segmentation algorithm directly determines the efficiency of the apple harvesting robot.

It is difficult to recognize green apples in the images of fruits and branches with similar background color characteristics, fruit occlusions, overlap, and various illumination conditions. Thus, the dataset is enriched as much as possible. Based on the actual environmental conditions, we continuously improve the proposed model to make it more suitable for apple segmentation and to meet the requirements of apple harvesting robots that work for all day conditions. Finally, the experiments show that the accuracy of the model is relatively high in complex orchard environments, especially in terms of the edge segmentation accuracy of the target fruits. The proposed method has demonstrated its potential to effectively improve the working efficiency of apple harvesting robots in real-world orchard environments. In addition, the computational time of our method is lower than that of other methods and our method basically meets the requirements of real-time segmentation of apple images. The segmentation performance of the proposed method is better than other methods, and the generalization ability of the model is strong. Although our method achieved relatively good results, there are still some areas that need to be improved. In the future, we need to increase the speed of the model and improve its efficiency without sacrificing segmentation accuracy.

Table 3

The specific result with Dice Rate, Recall Rate and Precision Rate (%).

Evaluation parameter	Covered fruit	Overlapping fruit	Single fruit
Dice Rate	94.46	93.58	97.15
Recall Rate	95.12	93.93	98.67
Precision Rate	96.23	95.78	98.32

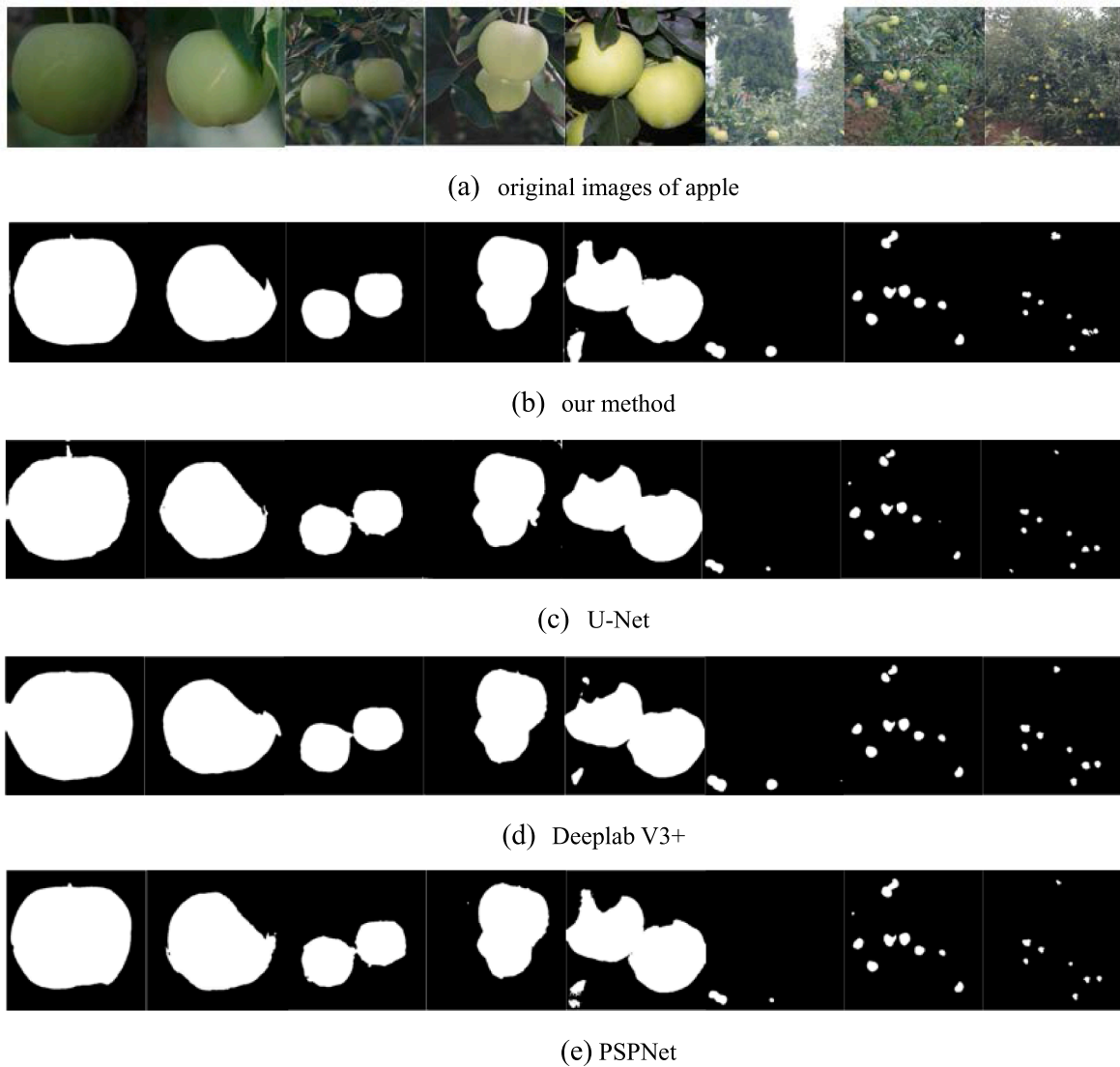


Fig. 8. Experiment result comparison.

Table 4

Comparison result of Dice (%).

experimental method	Covered fruit	Overlapping fruit	Single fruit
U-Net	91.90	90.03	94.57
Deeplab V3+	93.31	92.25	95.12
PSPNet	91.18	88.89	94.49
Ours	94.46	93.58	97.15

Table 5

Comparison result of recognition speed (s).

Experimental method	U-Net	Deeplab V3+	PSPNet	Ours
recognition speed	0.40	0.41	0.43	0.39

CRediT authorship contribution statement

Qianwen Li: Methodology, Software, Validation, Writing - original draft. **Weikuan Jia:** Conceptualization, Funding acquisition, Writing - review & editing. **Meili Sun:** Data curation, Writing - original draft. **Sujuan Hou:** Visualization, Investigation, Writing - review & editing. **Yuanjie Zheng:** Supervision, Funding acquisition.

Table 6

Comparison result of different data sets (%).

Parameter	Algorithm	Pascal Voc 2012	Cityscapes	Adobe's Portrait	Apple data set
Dice Rate	U-Net	87.24	86.52	93.89	92.16
	Deeplab V3+	89.01	88.59	94.21	93.56
	PSPNet	88.48	88.29	92.87	91.52
	Ours	91.11	90.61	95.21	95.11
	U-Net	89.31	87.82	94.01	93.23
Recall Rate	Deeplab V3+	91.64	91.73	94.48	94.81
	PSPNet	90.48	90.17	93.91	94.11
	Ours	93.32	93.49	96.89	96.23
	U-Net	90.24	90.02	94.87	95.08
	Deeplab V3+	92.48	91.91	94.85	95.19
Precision Rate	PSPNet	91.92	91.68	93.26	94.83
	Ours	94.56%	94.28	96.95	97.91

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by Focus on Research and Development Plan in Shandong Province (No.: 2019GNC106115); China Postdoctoral Science Foundation (No.: 2018M630797); National Natural Science Foundation of China (No.: 62072289); Shandong Province Higher Educational Science and Technology Program (No.: J18KA308); Taishan Scholar Program of Shandong Province of China (No.: TSHW201502038).

References

- Ahmad, M.T., Greenspan, M., Asif, M., et al., 2018. Robust Apple segmentation using Fuzzy Logic. 5th International Multi-Topic ICT Conference (IMTIC) IEEE, doi: 10.1109/IMTIC.2018.8467275.
- Bac, C.W., van Henten, E.J., Hemming, J., et al., 2014. Harvesting robots for high-value crops: State-of-the-art review and challenges ahead. *J. Field Rob.* 31 (6), 888–911.
- Bargoti, S., Underwood, J.P., 2017a. Image segmentation for fruit detection and yield estimation in apple orchards. *J. Field Rob.* 34 (6), 1039–1060.
- Bargoti, S., Underwood, J., 2017b. Deep fruit detection in orchards. *IEEE International Conference on Robotics and Automation (ICRA) 2017*, 3626–3633.
- Chen, L.C., Papandreou, G., Kokkinos, I., et al., 2017a. Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. *IEEE Trans. Pattern Anal. Mach. Intell.* 40 (4), 834–848.
- Chen, L.C., Zhu, Y., Papandreou, G., et al., 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation. *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 801–818.
- Chen, S.W., Shivakumar, S.S., Dcunha, S., et al., 2017b. Counting apples and oranges with deep learning: A data-driven approach. *IEEE Rob. Autom. Lett.* 2 (2), 781–788.
- Davidson, J., Silwal, A., Karkee, M., et al., 2016. Hand-picking dynamic analysis for underdense robotic apple harvesting. *Trans. ASABE* 59 (4), 745–758.
- Patrício, D.I., Riederer, R., 2018. Computer vision and artificial intelligence in precision agriculture for grain crops: A systematic review. *Comput. Electron. Agric.* 153, 69–81.
- Guo, Y., Liu, Y., Georgiou, T., et al., 2018. A review of semantic segmentation using deep neural networks. *Int. J. Multimedia Inform. Retrieval* 7 (2), 87–93.
- Hou, S., Zhou, S., Liu, W., et al., 2018. Classifying advertising video by topicalizing high-level semantic concepts. *Multimedia Tools Appl.* 77 (19), 25475–25511.
- Ji, W., Qian, Z., Xu, B., et al., 2019. Apple viscoelastic complex model for bruise damage analysis in constant velocity grasping by gripper. *Comput. Electron. Agric.* 162, 907–920.
- Ji, W., Zhao, D.A., Cheng, F.Y., et al., 2012. Automatic recognition vision system guided for apple harvesting robot. *Comput. Electr. Eng.* 38 (5), 1186–1195.
- Jia, W., Mou, S., Wang, J., et al., 2020a. Fruit recognition based on pulse coupled neural network and genetic Elman algorithm application in apple harvesting robot. *Int. J. Adv. Rob. Syst.* 17 (1), 897473.
- Jia, W.K., Tian, Y.Y., Luo, R., et al., 2020b. Detection and segmentation of overlapped fruits based on optimized mask R-CNN application in apple harvesting robot. *Comput. Electron. Agric.* 172, 105380.
- Jia, W.K., Zhang, Y., Lian, J., et al., 2020c. Apple harvesting robot under information technology: A review. *Int. J. Adv. Rob. Syst.* 17 (5), 925310.
- Kang, H., Chen, C., 2020. Fruit detection, segmentation and 3D visualisation of environments in apple orchards. *Comput. Electron. Agric.* 171, 105302.
- Koirala, A., Walsh, K.B., Wang, Z., et al., 2019a. Deep learning for real-time fruit detection and orchard fruit load estimation: Benchmarking of 'MangoYOLO'. *Precis. Agric.* 20 (6), 1107–1135.
- Koirala, A., Walsh, K.B., Wang, Z., et al., 2019b. Deep learning—Method overview and review of use for fruit detection and yield estimation. *Comput. Electron. Agric.* 162, 219–234.
- Kuruvilla, J., Sukumaran, D., Sankar, A., et al., 2016. A review on image processing and image segmentation. *Int. Conf. Data Mining Adv. Comput. IEEE* 198–203.
- Li, D.H., Zhao, H., Yu, X., 2019. Overlapping green apple recognition based on improve spectral clustering. *Spectroscopy and Spectral Analysis* 39 (9), 2974–2981.
- Liu, H., Liu, B., Zhang, H., et al., 2018a. Crowd evacuation simulation approach based on navigation knowledge and two-layer control mechanism. *Inf. Sci.* 436, 247–267.
- Liu, H., Xu, B., Lu, D., et al., 2018b. A path planning approach for crowd evacuation in buildings based on improved artificial bee colony algorithm. *Appl. Soft Comput.* 68, 360–376.
- Liu, X.Y., Jia, W.K., Ruan, C.Z., et al., 2018c. The recognition of apple fruits in plastic bags based on block classification. *Precis. Agric.* 19 (4), 735–749.
- Liu, X.Y., Zhao, D.A., Jia, W.K., et al., 2019. Fruits segmentation method based on superpixel features for apple harvesting robot. *Trans. Chinese Soc. Agric. Mach.* 50 (11), 15–23.
- Lv, J., Ni, H., Wang, Q., et al., 2019a. A segmentation method of red apple image. *Sci. Hortic.* 256, 108615.
- Lv, J.D., Wang, F., Xu, L., et al., 2019b. A segmentation method of bagged green apple image. *Sci. Hortic.* 246, 411–417.
- Majeed, Y., Zhang, J., Zhang, X., et al., 2020. Deep learning based segmentation for automated training of apple trees on trellis wires. *Comput. Electron. Agric.* 170, 105277.
- Rakun, J., Stajnik, D., Zazula, D., 2011. Detecting fruits in natural scenes by using spatial-frequency based texture analysis and multiview geometry. *Comput. Electron. Agric.* 76 (1), 80–88.
- Ronneberger, O., Fischer, P., Brox, T., et al., 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. *Med. Image Comput. Comput.-Assisted Intervention-MICCAI* 234–241.
- Silwal, A., Davidson, J.R., Karkee, M., et al., 2017. Design, integration, and field evaluation of a robotic apple harvester. *J. Field Rob.* 34 (6), 1140–1159.
- Sui, X., Zheng, Y., Wei, B., et al., 2017. Choroid segmentation from optical coherence tomography with graph-edge weights learned from deep convolutional neural networks. *Neurocomputing* 237, 332–341.
- Sun, S., Jiang, M., He, D., et al., 2019. Recognition of green apples in an orchard environment by combining the GrabCut model and Ncut algorithm. *Biosyst. Eng.* 187, 201–213.
- Takikawa T, Acuna D, Jampani V, et al. 2019. Gated-scnn: Gated shape CNNs for semantic segmentation. *Proceedings of the IEEE International Conference on Computer Vision*. 5229–5238.
- Tripathi, M.K., Maktedar, D.D., 2020. A role of computer vision in fruits and vegetables among various horticulture products of agriculture fields: A Survey. *Inform. Process. Agric.* 7 (2), 183–203.
- Wang, D., Xu, Y., Song, H., et al., 2015. Fusion of K-means and Ncut algorithm to realize segmentation and reconstruction of two overlapped apples without blocking by branches and leaves. *Trans. Chinese Soc. Agric. Eng.* 31 (10), 227–234.
- Wang, D.D., He, D.J., Song, H.B., et al., 2019. Combining SUN-based visual attention model and saliency contour detection algorithm for apple image segmentation. *Multimedia Tools Appl.* 78 (13), 17391–17411.
- Xu, W., Chen, H., Su, Q., et al., 2019. Shadow detection and removal in apple image segmentation under natural light conditions using an ultrametric contour map. *Biosyst. Eng.* 184, 142–154.
- Zhang, J., Ding, S.F., Jia, W.K., 2020. An adversarial non-volume preserving flow model with Boltzmann priors. *Int. J. Mach. Learn. Cybern.* 11 (4), 913–921.
- Zhao H, Shi J, Qi X, et al. 2017. Pyramid scene parsing network. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2881–2890.